9

Christian anthropology, philosophy, and science

1 Introduction

There is a list of topics regarding human nature that have been debated by apologists, theologians, philosophers, and others since the beginning of the development of the Christian Tradition; for example, the meaning of our having been created in the image of God, the sources and consequences of sin, the means by which Jesus has redeemed us, our fate after death. This chapter will concentrate on only two questions: the metaphysical composition of humans (stated most briefly, dualism versus monism) and the closely related question of what happens after death.

In line with the general focus of this text, the *potential* conflict between traditional Christian views of dualism and current neurobiological research suggesting a physicalist view of humans will play a large part, in that *if* this turns out to be an *unavoidable* conflict it has implications that would constitute a significant crisis for the Christian Tradition. It is for this reason that I shall give much more attention to physicalism than dualism in this chapter.

This chapter will begin (in section 2) with the task of sorting through the history of Christian views of human nature. Section 3 presents a small sampling of results from neuroscience that strongly suggest a physicalist view, in that human capacities previously attributed to the soul (or mind)[1] can be shown to depend on quite specific locations or systems in the brain.

While Christian acceptance of a physicalist account of human nature raises, or brings to the fore, a number of philosophical issues, I shall concentrate, in section 4, on the problem of reductionism: that is, how can it *fail to be the case* that all beliefs and actions are simply determined by the operation of our brains. I claim that the answer involves a striking worldview change that is part of the separation of modernity from whatever comes next in Western intellectual history.

2 A patchy history of Christian views of human nature

A number of years ago my colleague in neuropsychology, Warren Brown, warned that a new conflict between Christianity and science, comparable to that between

[1] Recall that during part of Christian history these terms could be used interchangeably.

creationism and evolutionism, was on the horizon. Most Christians, he said, were substance dualists, while developments in the cognitive neurosciences were strongly suggesting that humans are all of a piece and that our immensely complex brains, conditioned by our immensely complex cultures, were the source of higher human capacities. Some philosophers and scientists were already claiming that neuroscience falsified the beliefs of countless religious people. Brown suggested that we produce a book showing that there is an acceptable Christian account of human nature that is consonant with neuroscience. Recall my mention (in Chapter 8) of Ernan McMullin's concept of *consonance* as the ideal relation between theology and science.

We published chapters by scientists from a variety of relevant disciplines, as well as a theologian, a biblical scholar, and a Christian ethicist, all supporting the position best known in philosophy as "nonreductive physicalism."[2] I had two jobs: One was to write a chapter providing a philosophical argument *against* the reduction of all human capacities and behavior to mere brain processes. I do not judge that chapter to have been successful.[3]

My second job was to write an Introduction to the book incorporating a historical account of Christian views on this particular aspect of Christian anthropology. I found it surprisingly difficult. It made sense to begin with the Bible. Not being a biblical scholar, I resorted to reference works, but concluded that the views *attributed* to biblical authors turned out to be strangely correlated with predominant views of human nature held at the time the reference works were written.

I next attempted to piece together the transition from what appeared to be a monistic Hebraic account in the Bible (to be explained below) to a Platonic-inspired body–soul dualism espoused by Augustine. This, too, proved confusing, as I shall explain below (section 2.2). Much has already been written here (Chapter 1) on the medieval period, in which Aristotelian influences competed with the predominantly Platonic–Augustinian tradition. Section 2.3 will review and amplify contributions from modern philosophy to Christian anthropology and attempt to do justice to contemporary developments.[4]

[2] Warren S. Brown, Nancey Murphy, and H. Newton Malony, eds., *Whatever Happened to the Soul? Scientific and Theological Portraits of Human Nature* (Minneapolis: Fortress Press, 1978).

[3] So please do not rely on it in any of your own work. See instead Nancey Murphy and Warren S. Brown, *Did My Neurons Make Me Do It? Neurobiological and Philosophical Perspectives on Moral Responsibility and Free Will* (Oxford: Oxford University Press, 2007).

[4] Since my early attempts to find authoritative and comprehensive accounts of Christian developments in doctrines of human nature, I have discovered Raymond Martin and John Barresi, *The Rise and Fall of Self and Soul: An Intellectual History of Personal Identity* (New York: Columbia University Press, 2006), which covers biblical sources, philosophical sources, and developments up to the present, including those suggested by scientific advances.

2.1 The intriguing tale of human nature in the Bible

While Brown was correct regarding the prevalence of dualism among Christians today, there is another position, nearly as popular, called "trichotomism." This is said to be based on Paul's blessing in 1 Thessalonians 5.23: "May the God of peace himself sanctify you entirely; and may your spirit and soul and body be kept sound and blameless at the coming of our Lord Jesus Christ" (NRSV). However, a more proximate source seems to be the writings of Chinese evangelist Watchman Nee. So there are three major positions among contemporary Christians on the biblical view of human nature. However, a book review claims that 130 different views of the human person have been documented.[5]

An important question for contemporary Christians, then, is whether it is possible to settle differences among monists, dualists, and trichotomists, and if so, how? According to Brown, it is necessary to seek consonance among five sources: Scripture, tradition, science, reason, and human experience. I interpret "reason" to stand largely for philosophy. The structure of this section (as well as of the rest of the chapter) will be heavily indebted to Brown's proposal.

2.1.1 The Bible on monism, dualism, or . . .?

For over 16 centuries the predominant Western Christian view has been that the Bible, both OT and NT, taught some form of substance dualism. This consensus began to be called into question very early in the twentieth century. I have found it important to point out the long history of this shift in Christian thinking in order to try to head off objections that my anti-dualism (and that of other Christians) fits the old canard of theology challenged by science, and theologians then backing down.

I believe that, a century later, there is coming to be a consensus that the Bible does not teach (or even presuppose) any sort of substance dualism. Systematic theologian Veli-Matti Kärkkäinen writes that "a number of more recent, extensive studies have led to verdicts . . . supportive of Paul's essential wholism" and "emphasis on embodied life in this world and the next, while combating body-soul dualism."[6] Nevertheless, Kärkkäinen believes that the term "soul" can and should continue to be used in theology because of its prevalence in the tradition and because it is the theologian's job to help the faithful grasp its redefinition.[7]

[5] Review by Graham McFarlane of N. H. Gregersen et al., eds., *The Human Person in Science and Theology* (Edinburgh: T. & T. Clark, 2000), in *Science and Christian Belief* 14, no. 1 (April 2002): 94–5.

[6] Veli-Matti Kärkkäinen, *Creation and Humanity: A Constructive Christian Theology for the Pluralistic World*, vol. 3 (Grand Rapids, MI: Eerdmans, 2015), 333–4; quoting Joel B. Green, *Body, Soul, and Human Life: The Nature of Humanity in the Bible* (Grand Rapids, MI: Baker Academic Press, 2008), 7–8.

[7] Kärkkäinen, *Creation and Humanity*, 345. However, my advice is that it would be better simply to "put the word to bed" for about 30 years, based on my experience that some terms carry so much unhelpful baggage that it is difficult for people even to notice that a familiar term has been redefined.

Joel Green (b.1956) points out that modern thinkers need to recognize that many of the questions we pose were of no interest to biblical authors.[8] Thus, in some cases the questions may turn out to be impossible to answer. The pertinent example is that our contemporary conceptions of human nature are strikingly different from those of the Hebraic tradition.

I describe modern people in the West as thinking of themselves as self-contained in two senses: First, modern individualism, taken over from early modern atomism in physics, assumes that the individual has ontological priority over social groups—that is, I could be the person I am regardless of my social interactions. Second, we have the metaphor of the true self being contained inside the body—a metaphor universalized by René Descartes, but a likely import from the Catholic spiritual tradition, going back to Augustine (see Chapter 2, section 4.1). Philosophical theologians Fergus Kerr and Nicholas Lash describe this as the little person in the head. Phillip Cary emphasizes the modern belief that the little person can never get out.

This is in contrast to interpretations of OT anthropology, such as that of Aubrey Johnson (1901–95): the Hebraic personality was thought to be extended in subtle ways among the community by means of speech and other communication. "Accordingly, in Israelite thought the individual, as a . . . centre of power capable of indefinite extension, is never a mere isolated unit."[9]

One of the questions that has, until recently, seemed impossible to answer has been the long-standing debate regarding Paul: did he hold to the monism of his Hebrew tradition or accept the dualism of Hellenistic philosophers, already prevalent in his context for several centuries?

James Dunn (b.1939) proposes that, while Paul was clearly familiar with both, he found it necessary to craft an account of human nature that was adequate to his theology regarding the impact of divine revelation and grace on the human being.[10] He needed a conception of human life *now* that could support the promise of resurrection. How can we understand ourselves in a way that we can speak of our embodied selfhood in this life, while imagining a new kind of embodiment raised to incorruptibility, glory, and power, patterned on the model of Christ's resurrected body?[11]

For two reasons I have found a distinction of terms used by Dunn particularly useful for understanding biblical anthropology, and also for explaining how

[8] Joel B. Green, *Body, Soul and Human Life: The Nature of Humanity in the Bible* (Grand Rapids, MI: Baker Academic, 2008), 12–13; following Klaus Berger, *Identity and Experience in the New Testament* (Minneapolis: Fortress, 2003), 6.

[9] Aubrey R. Johnson, *The One and the Many in the Israelite Conception of God*, 2nd ed. ([1942] Cardiff: University of Wales Press, 1961), 4. See also Robert A. Di Vito, "Old Testament Anthropology and the Construction of Personal Identity," *Catholic Biblical Quarterly* 61, no. 1 (1999): 217–38.

[10] James D. G. Dunn, *The Theology of Paul the Apostle* (Grand Rapids, MI: Eerdmans, 1998), 51.

[11] Dunn, *Paul*, 60.

it could have been possible for Christians, for so many centuries, to believe that the Bible required substance dualism. Dunn distinguishes between what he calls "aspective" and "partitive" accounts of human nature. He writes:

> in simplified terms, while Greek thought tended to regard the human being as made up of distinct parts, Hebraic thought saw the human being more as a whole person existing on different dimensions. As we might say, it was more characteristically Greek to conceive of the human person "partitively," whereas it was more characteristically Hebrew to conceive of the human person "aspectively." That is to say, we speak of a school *having* a gym (the gym is part of the school); but we say I *am* a Scot (my Scottishness is an aspect of my whole being).[12]

Thus, many ancient Greek philosophers were interested in the question: what are the essential parts of a human being? In contrast, for the biblical authors each "part" ("part" in scare quotes) stands for the whole person thought of from a certain angle. For example, Paul's distinction between spirit and flesh is not our later distinction between soul and body; "spirit" stands for the whole person in relation to God. What the NT authors are concerned with, then, is human beings in relationship to the natural world, to the community, and to God. Paul is concerned with two ways of living: one in conformity with the Spirit of God, and the other in conformity to the old eon before Christ.

A second factor in confusions regarding human nature, and perhaps contributing to partitive readings of biblical texts, is the fact that the Scriptures used by early Christian thinkers were those that Christians call the OT. What early Christian apologists had available to them, especially apologists to the world outside of traditional Hebraic Judaism, was the *Septuagint*, a translation of the Hebrew texts into Greek, said to have been done by 70 scholars (thus the name), and completed around 250 BC. Given the vastly different worldviews from which these two languages derived (I might even say, incommensurable worldviews), it was sure to be the case that Greek thought would have entered the texts because of lack of available terminology in Greek to capture the meaning of a Hebrew concept in a single word. For example, one of the words used frequently in the OT, *nephesh*, derived originally from its designation of the throat, was extended to the breath that flowed through the throat, and extended even further to the life force, since the dead no longer breathe. This term was translated into Greek as *psyche*. Thus, early English versions used "soul" almost exclusively to translate it. Newer translations substitute a variety of terms, such as "living being." Even translations made by scholars with a more conservative point of view show a shift toward replacements of "soul" with other, more appropriate, terms.

[12] Dunn, *Paul*, 54.

2.1.2 Biblical views of the afterlife

In Jesus' day there were three accounts of what happens at death. One was the predominant view of earlier Hebrew texts: death is simply death, and hope for the future was hope for the future of Israel. Joel Green, following a number of noted OT scholars, notes that there is no well-defined account of what happens after death.[13] However, Jewish scholar Jon Levenson (b.1949) writes that the concept of the dead residing in Sheol reflects an impressively long-standing and broad consensus. The dead do not cease to exist altogether nor are they entirely oblivious, but it is a negative experience.[14] It is the destination of all, both good and bad.

If Levenson is correct here, there are resemblances to numerous other accounts of the netherworld, and the doctrine of Sheol may have been influenced by other ancient cultures in the region. Levenson, in fact, mentions Homer. Recall (from Chapter 1, section 2.1) Paul Feyerabend's account of the incommensurability between Homeric and classical Greek thought. Homeric humans were simple additions of parts: body parts and mental parts; the latter not coming from a unified, invisible, interior entity. Levenson notes one translator of the Odyssey who describes the netherworld as "this gloom at world's end."[15]

The second account of the afterlife in Jesus' (and Paul's) day was resurrection. Acts 23.8 makes clear that there was a major controversy between the Sadducees (who rejected it) and the Pharisees (who accepted it). While resurrection only became widely accepted during Second Temple Judaism (515 BC – AD 70), Levenson writes that there are hints of it in the Torah, such as in Psalm 40.2: "He drew me up from the desolate pit, out of the miry bog, and set my feet upon a rock, making my steps secure" (RSV). However, it was thought of only as an unlikely gift from God.[16] The concept of resurrection is usually taken to be clearly endorsed much later in Daniel. Both Levenson and another contemporary Jewish scholar, Neil Gillman (1933–2017), argue that resurrection represents the only authentic Jewish eschatology.[17] Of course, for Christians, the resurrection of Jesus played a great role. The peculiarity for Jewish Christians was that it was expected only at the end of history.

The third option during Jesus' lifetime, for those who had adopted body–soul dualism, was that the soul would continue to exist after the death of the body. For many, the escape of the soul from the body was highly desirable. Perhaps one reason for difficulties in interpreting Paul's second letter to the Corinthians is that

[13] Green, *Human Life*, 152–7.
[14] Jon D. Levenson, *Resurrection and the Restoration of Israel: The Ultimate Victory of the God of Life* (New Haven, CT: Yale University Press, 2006), 35.
[15] Levenson, *Resurrection*, 37; citing Robert Fitzgerald.
[16] Levenson, *Resurrection*, 37.
[17] Neil Gillman, *The Death of Death: Resurrection and Immortality in Jewish Thought* (Woodstock, VT: Jewish Lights Publishing, 2000).

he was attempting to write to three audiences at once: those who had no expectation of life after death, those who believed in bodily resurrection but did not have a clear understanding of Jesus' resurrection, and Hellenized Christians who did not understand why returning to bodily existence would constitute good news. Later Christians created a composite account of life after death: the body dies, the soul lives on, and is later reclothed in a resurrection body.

2.1.3 So that what is mortal may be swallowed up in hope[18]

Notice that when Christians focus on resurrection as the hope of what is to come, the old storyline from perfect creation, to fall, to restoration must yield to a storyline from creation, through slow and painful development, from the simple and chaotic to the complex and orderly. But then what? Christian hope for the future is based not on freeing the immortal soul, but rather on resurrection. And the only vision of the end of the world that is fully consistent with the hope of resurrection is a transformation of the whole cosmos, a transformation of which Jesus' resurrection is first fruits. We can say nothing of what this transformation will be like in scientific terms because science is based on the way things are in this eon. But we can say much about the new world in moral terms. It will be a world whose character Isaiah evoked in his prophecy:

> For I am about to create new heavens and a new earth . . . I am about to create Jerusalem as a joy, and its people as a delight . . . No more shall the sound of weeping be heard . . . or the cry of distress. They shall build houses and inhabit them; they shall plant vineyards and eat their fruit . . . Before they call I will answer, while they are yet speaking I will hear. The wolf and the lamb shall feed together, the lion shall eat straw like the ox. They shall not hurt or destroy on all my holy mountain, says the Lord. (Isa. 65.17–25 NRSV *passim*)

If we reject the Neoplatonic vision of the "flight of the alone to the Alone," and return to the biblical view of the rule of God "on earth as it is in heaven," we find a vision of the end time that shows the ultimate value of sociality; that shows that history is meaningful, because past achievements are not left behind but transformed, past sorrows add poignancy to present joy. Finally, it is a vision that shows there to be ultimate value in our care for and harmony with the whole of nature.

2.2 Human nature in the tradition

In this subsection, I try to do justice to ancient and medieval anthropology, to Reformation developments, and to modern and contemporary thought. The last brings us, in a sense, full circle in that contemporary theology and biblical studies

[18] 2 Cor. 5.14.

are powerfully shaped by the recovery of a monist anthropology and the centrality of the resurrection.

2.2.1 Ancient and medieval positions

It has been common to describe the demise of dualism and accompanying immortality of the soul as occurring during the first half of the twentieth century: it was seen as Greek corruption of original Hebraic Christianity, with its emphasis on resurrection of the body. However, Green regularly points out that the sharp distinction between Greek dualism and Hebraic monism is an oversimplification. I now know this all too well. In an attempt to give an account of the development from post-NT monism to the Platonic-inspired dualism of Augustine, I discovered that there are a vast number of positions, and that they do not fit into any detectable chronological order, except to say that there was a *general* progression from monism to dualism and toward the addition of a concept of immortality.[19]

I should have expected this, given my emphasis on the need to interpret essential terms in the context of the various authors' overall positions. These positions were developed in response both to a variety of philosophical systems (Platonism and Neoplatonism, but also Stoicism and atomism), and to a variety of religious competitors such as Gnosticism and Manichaeism.

One difficulty for contemporary interpreters lies in the fact that ancient philosophy was still in the process of defining materiality, and it was difficult, then, to imagine the non-material, except in Platonic Forms. Thus, we too easily read into early accounts the radical dualism of material versus non-material. (We should be struggling, ourselves, to redefine the material in light of quantum physics.)

Here are a few examples of uses of words in ancient documents that are at variance with current expectations. (1) Resurrection: In contrast to delicate attempts to read NT accounts of resurrection in terms of a physical *and transformed* body, it could later be understood in a purely spiritualist manner, or, more often, as a mere reconstruction of the physical body. For dualists it can mean resurrection of both body and soul. (2) Dualism: It can be understood entirely in materialist terms; that is, both body and soul are composed of matter. However, there were also instances of a more Platonic account of material body and non-material soul. (3) Immortality: It can be used interchangeably with "resurrection." It can be seen either as a gift of God or as a natural endowment of the soul. (4) The fate of the damned: It can be simply death. Hell can be understood as a quick means of destruction, or as a means of eternal punishment.

[19] See Caroline Walker Bynum, *The Resurrection of the Body in Western Christianity, 200 – 1336* (New York: Columbia University Press, 1995) for a detailed account of varying concepts of resurrection and associated anthropologies. Another valuable resource is Raymond Martin and John Barresi, *The Rise and Fall of Soul and Self: An Intellectual History of Personal Identity* (New York: Columbia University Press, 2006).

By Augustine's day, these issues were resolved, and his position continued to influence Christian thinking throughout the Middle Ages. Chapter 1 gave a great deal of attention to the influences of ancient philosophy on major theologians: Augustine and Thomas. However, the emphasis there was on the effects of Greek philosophy on the concepts of reason that they brought to their theologies. I shall mention here a bit more about philosophical influences on their accounts of human nature. Pythagoras is thought to have originated the doctrines of the tripartite soul and of reincarnation, which surfaced again in Plato's thought.

Augustine retained the Platonic notion of the tripartite, immaterial soul, but rejected Plato's view of the soul as eternal, replacing it with mere immortality. He also altered Plato's account of the parts themselves. For Plato, the intellect was the highest level. The intermediate level, *thumos*, is often translated as "spirit" but it is not spirit in the Christian sense, but something closer to what we mean in speaking of a "spirited" horse; and the lowest was appetite. It is said that Augustine invented our Western concept of the will, and for him it was the highest level, followed by intellect and appetite. The Neoplatonism of the early Christian era (Chapter 1, section 4) also contributed to Augustine's understanding of human nature.

Thomas adopted Aristotle's concept of the soul as the immanent, substantial Form of the body, giving the body all of its characteristic capacities. He agreed with Aristotle in attributing souls to all living things: the vegetative soul to plants, the sensitive soul to animals, and the rational soul to humans. These are in a sense cumulative: we share with plants and animals the levels of soul that mirror theirs. However, Thomas had the difficult task of arguing, against Aristotle, that the rational soul could survive death of the body. He did not see this as an ideal state, since embodiment, along with all of the other faculties of the soul, are needed in order to function as a complete human being. Thus, resurrection of the body was essential.

As noted (in Chapter 1), Thomas received Aristotelian thought through Islamic intermediaries. I believe that, with their amplifications, Thomas's account of the soul is the most elaborate and interesting to be found in Christian history. I shall not describe it further here, since I use his list of its capacities in section 3 below.

At first, some of the ideas found in Thomas's works were censured by the church, but he was canonized in 1323, and his teachings have been regarded as a touchstone of Catholic thought, especially recognized at the Council of Trent (1545–63). Nonetheless, there were a variety of contrasting conceptions of human nature throughout the medieval period (see Chapter 6, section 3.2.1 on the Spiritualist Tradition to get some idea of the variety).

2.2.2 Reformation reconfigurations

Here I shall distinguish between the well-known Magisterial Reformation, and the lesser-known Radical Reformation. As throughout this chapter, I shall restrict my

account to the metaphysical make-up of humans and their fate after death. I shall further restrict my treatment of the Magisterial Reformers to the two most influential: Martin Luther (1484–1546) and John Calvin.

A striking feature, in my view, is that so little can be found in the secondary literature on theories of human nature during this period. Perhaps this should not come as a surprise because the pressing issues had to do, in the case of humans, with sin and salvation. The broader issue, of course, was the question of the authority of the church. However, the principle of *sola Scriptura* meant that any doctrines regarding human nature and the afterlife were supposed to be based on Scripture alone. This certainly required rejecting Thomas's Aristotelian account. However, I believe that by the time of the Reformation it had come to be nearly impossible to read Scripture except in partitive terms, and both Luther and Calvin believed that Augustine had already formulated the biblical view of human nature.

Given so much agreement regarding Augustine's grasp of biblical teaching, there remained few issues separating Luther and Calvin. One was the question of whether the soul was a special creation of God, endowed at conception or, whether it, like the body, is produced by propagation (traducianism). Although Augustine "was inconclusive" on this issue, Luther believed that he leaned toward traducianism, and therefore he accepted it.[20] Calvin was a creationist.

The two Reformers also disagreed on what happens at death. They both believed in resurrection and the existence of the soul. But what happens to the soul between the death of the earthly body and the general resurrection? This is the question of the intermediate state. Are the souls of the just enjoying conscious awareness of God (Calvin's position) or are they "sleeping"? However, because "sleep" is sometimes used as a euphemism for death in the Bible, there were actually two possibilities, either that the soul dies with the body or that it (more literally) sleeps. Luther compares soul-sleep to normal sleep: just as one falls asleep and unexpectedly awakes, the dead, when they rise, will not know how they have come into and through death.

It is clear that interest in this question grew out of Luther's and his followers' objections to the Catholic Church's doctrine of purgatory, for neither position allows that those destined to be saved could suffer through an intermediate period of purifying suffering.

The Radical Reformation, being a highly variegated and decentralized movement, is difficult to date, but perhaps its clearest representative was the Swiss Brethren, who separated from Huldrych Zwingli (1484–1531) and his reformation in Zürich in 1525. Their objection was Zwingli's delaying of reforms to suit

[20] George Huntston Williams, *The Radical Reformation*, 3rd rev. and expanded ed. ([1962] Kirksville, MO: Sixteenth Century Journal Publishers, 1992), 793.

the judgment of the city council. After the Swiss Brethren went their own way, they strove to complete a return to NT Christianity, and in the process rejected infant baptism, not only because they believed it was not a biblical practice, but also because it had become the essential marker for citizenship. Thus, they began to baptize themselves as believers, and thereafter were horribly persecuted (as Anabaptists—re-baptizers) both for heresy and for civil disobedience.

I first became interested in Radical Reformation anthropology due to reading about the intermediate state; it was reported that in addition to Luther, many Anabaptists believed in soul-sleep.[21] I assumed then, as many do, that there must *be* a soul that sleeps, and therefore that Anabaptists, too, were (partitive) body–soul dualists. However, this was before learning that "soul-sleep" could also refer to death of the soul along with the body, and also, especially, before I encountered Dunn's partitive/aspective distinction. I have now found evidence that many Radical Reformers were aspective monists.

My first clues came from an issue of a journal, *Direction: A Mennonite Brethren Forum*, which had devoted a number of articles to the physicalism/dualism debate.[22] I found the article by Terry Hiebert suggestive. He states that his general purpose in his review article is to try to answer the question (exacerbated by my book, *Bodies and Souls*) of whether the soul actually exists. More particularly, he says, "it is time to reconsider the admitted paradoxes surrounding Anabaptist views of the soul and death."[23]

Hiebert displays a remarkable breadth of knowledge regarding the Anabaptist tradition. He sketches the positions on the soul and afterlife of three of the most important founders: Michael Sattler (*c.*1490–1527), Balthasar Hubmaier (*c.*1480–1528), and Menno Simons (*c.*1496–1561), stating that they represent a spectrum of views on the nature of the soul and its implications for the afterlife.

> Sattler and Menno refused to speculate on the nature of the soul, but referred to the soul as a believer's openness to God. Flesh and spirit were terms more suitable for describing the Christian life of separation and holiness . . . Sattler expected to die and wait for the resurrection. Hubmaier expected his [immortal] soul to meet God at death and Menno expected to rest in Paradise until the seed of the new body was born at the resurrection.[24]

[21] As have other Christian bodies, such as the Quakers, recent descendants of the Radicals have adopted the intentionally pejorative name originally given to them.

[22] Terry G. Hiebert, "Is the Search for the Anabaptist Soul a Dead End? Historic Anabaptism Meets Nancey Murphy's Nonreductive Physicalism," *Direction: A Mennonite Brethren Forum* 47, no. 2 (Fall 2008): 185–200. Here he focuses almost entirely on my *Bodies and Souls or Spirited Bodies?* (Cambridge: Cambridge University Press, 2006).

[23] Hiebert, "Anabaptist Soul," 187.

[24] Hiebert, "Anabaptist Soul," 189.

Later in the article he expresses what I take to represent the "paradoxes" surrounding Anabaptist views of the soul:

> First, historic Anabaptist writers and church confessions were remarkably unified in using the biblical language of the soul. While more research is needed, apparently most Anabaptists, reflecting biblical diversity, were either implicit or explicit dualists. However, their concerns for holiness and discipleship inclined them to talk about the soul in non-speculative and pastoral terms. Anabaptists practiced a restrained use of this interior language of the soul in worship and ministry. They believed in the soul's engagement with God and people through baptism, the Lord's Supper, footwashing, marriage, and pastoral care. *This rich metaphor evokes the moral, emotional, relational, volitional, rational, and spiritual dimensions of human experience.* With the exception of the most explicit dualists, Anabaptists could have affirmed that we *are* souls rather than *have* them.[25]

These long quotations illustrate the value of the aspective/partitive distinction. Apart from Hubmaier, who was an explicit dualist (in fact, a trichotomist), it seems that there is no paradox regarding the (early) Anabaptist concept of the soul so long as it is read aspectively rather than partitively.

A final note on the Reformation: at the Council of Trent, instrumental in the Catholic "counter-reformation," the wakeful sleep of the dead during the intermediate state was affirmed as doctrine.

2.2.3 Modern and contemporary developments

It would take an entire chapter to do justice to theological developments after the Reformation. What can be said, briefly, is that most theologians continued to assume the dualistic anthropology of Catholicism and the Magisterial Reformers into the twentieth century. It also influenced some later Anabaptist theologians. The eschatology that combined an immortal soul with bodily resurrection also continued.

In the seventeenth century, science and philosophy had an impact. The new atomism in physics marked an important turning point for theories of human nature. That is, if the hylomorphic worldview was rejected, then so too must be the Scholastic doctrine that the soul is the Form of the body.[26] There were two major options, which have survived in various forms until today.

As already noted, the Magisterial Reformers had rejected Aristotelian thought in the sixteenth century and returned to the more Platonic dualism of Augustine. Descartes, although a Catholic, chose this route, but created a dualism more radical

[25] Hiebert, "Anabaptist Soul," 197; first italics mine.

[26] Nonetheless, many Catholic thinkers still hold this view.

than Plato's.[27] Descartes distinguished between material (extended) substance and thinking substance; the body was the former, the mind the latter. Much of what we now call the philosophy of mind involved an attempt throughout the modern period to explain how these two could interact. Another now familiar influence of Descartes's writings was his importation of Augustine's peculiar form of inwardness from the Catholic spiritual tradition into modern philosophy (see Chapter 2, section 4.1). Fergus Kerr claims that even contemporary theologians who criticize Descartes's philosophy still exhibit the effects of his "inside-out" philosophy in their theological methods.[28]

The second obvious option was to postulate that humans are purely physical. Even though Thomas Hobbes used the word "mind," he believed that thinking was a matter of constructing useful strings of words, and that words represented physical things. Particles striking the senses could have further effects in the heart, and these are called passions. Human bodies, like all bodies, are entirely pushed by forces such as attraction, repulsion, and fear. He believed in souls, angels, and God, but believed that they too were entirely material. Although early church fathers such as Irenaeus[29] and Tertullian (*c.*150–225) had been materialists, dualism had taken such a strong hold on Christianity in later years that it was Cartesian dualism rather than Hobbes's materialism that influenced most Christian thought.

Historical criticism of the Bible has had a major impact on modern conceptions of human nature, but there have been contradictory tendencies. In the eighteenth and especially nineteenth centuries, many NT scholars cast doubt on the historicity of miracles in general and especially the great miracle of Jesus' resurrection. Skepticism about Jesus' resurrection led to increased emphasis among theologians on immortality of the soul as the only basis for Christian hope in the afterlife. Consider Adolf von Harnack's (1851–1930) neat summary of the kernel of Christian doctrine: The fatherhood of God, the brotherhood of man, and the infinite value of the human *soul*.[30]

Meanwhile—and here is the contradictory tendency—biblical scholars had begun to question whether body–soul dualism was in fact the position to be found in Scripture. One important contribution here was the work of H. Wheeler

[27] Green has mentioned on a number of occasions that contemporary readers often read Plato's writings through Cartesian eyes. I add that what it even means to be material was one of the central issues of ancient philosophy.

[28] Fergus Kerr, *Theology after Wittgenstein*, 2nd ed. ([1986] London: SPCK, 1997).

[29] Irenaeus' dates are surprisingly uncertain; his birth is thought to be either between 115 and 125 or between 130 and 142. His death is *c.*200–3.

[30] Adolf von Harnack, *Das Wesen des Christentums* (1900); translated as *What is Christianity?* (1901); my italics.

Robinson (1872–1945), an OT scholar, who argued that the Hebrew idea of personality is that of an animated body, not (like the Greek) that of an incarnated soul. However, Robinson held to an Idealist philosophy (all of reality is, in some sense, mental), so this was the beginning of a monist, but not physicalist, interpretation of the OT. He also believed that the NT had made a great advance in the belief that the essential personality (soul or spirit) survived death.

A decisive contribution was Rudolf Bultmann's claim in his *Theology of the New Testament* that Paul uses *soma* ("body") to characterize the human person as a whole.[31] In 1955 Oscar Cullmann (1902–99) gave the lectures that were published as *Immortality of the Soul or Resurrection of the Dead: The Witness of the New Testament*. Cullmann effectively contrasted biblical attitudes toward death, along with expectation of bodily resurrection, with Socrates' attitude, given his expectation that his soul would survive the death of his body.[32]

So here we have, in a sense, come full circle to the account of biblical anthropology above (section 2.1), with the mid-twentieth-century emphasis on a holist account of human nature and an emphasis on bodily resurrection. And although we have also seen that the accepted distinction between the Hebraic and Greek accounts of human nature was too sharply drawn, contemporary theology tends toward a more nuanced monism.

I had the opportunity to teach a doctoral seminar on contemporary theological anthropology with my theology colleague at Fuller Seminary, Veli-Matti Kärkkäinen. He rejects dualism and says that the best current theologians do so also. But nonetheless he disliked the term "nonreductive physicalism" because it seemed to emphasize the physical at the expense of all our other characteristics. Another term sometimes used for a non-dualist position is called "dual-aspect monism," meaning that we are unified beings, but we have both physical and spiritual aspects. But this distinction of physical versus spiritual still seems too simplistic. Based on the complex account Scripture gives of the various aspects of human life, we decided that the best term for Christian theology would be "multi-aspect monism." To illustrate, and for a fitting conclusion to this section, I quote Dunn's summary of Paul's conception of human nature:

> Paul's conception of the human person is of a being who functions within several dimensions. As embodied beings we are social, defined in part by our need for and ability to enter into relationships, not as an optional extra, but as a dimension of our very existence. Our fleshness attests our frailty and weakness as mere humans, the inescapableness of our death,

[31] Rudolf Bultmann, *Theology of the New Testament*, vol. 1 (New York: Scribner, 1951).

[32] Oscar Cullmann, *Immortality of the Soul or Resurrection of the Dead?* (New York: Macmillan, 1958).

our dependence on satisfaction of appetite and desire, our vulnerability to manipulation of these appetites and desires. At the same time, as rational beings we are capable of soaring to the highest heights of reflective thought. And as experiencing beings we are capable of the deepest emotions and the most sustained motivation. We are living beings, animated by the mystery of life as a gift, and there is a dimension of our being at which we are directly touched by the profoundest reality within and behind the universe. Paul would no doubt say in thankful acknowledgement with the psalmist: "I praise you, for I am fearfully and wonderfully made" (Ps. 139.14).[33]

3 Neuroscience and human nature

This section's purpose is to convey some of the striking correlations between human capacities that have been attributed to the soul (or mind) and highly specific regions or systems in the brain. It is tempting to say that these brain processes are in fact responsible for our "soulish" capacities. However, Warren Brown is correct in saying that a mental process is a *contextualized* brain event—contextualized in the body, with its history of interactions with the world, and especially, for higher human capabilities, contextualized in the social world with its sometimes centuries-old cultural history.

Although I do not have any concept of the soul, and, as you have read above, there is a vast number of such concepts, the one I find most interesting is Thomas's. This is not because of his metaphysical views on what the soul is,[34] but because he and his predecessors, both Aristotle and Muslim commentators on Aristotle's position, were such excellent observers of humans and other organisms. So I shall use Thomas's list of *capacities* to structure my exposition of neuroscientific findings.

3.1 Varied human capacities attributed to Thomas's soul

We have already seen that both Aristotle and Thomas attributed souls to plants and animals as well as humans. The human soul is tripartite, incorporating all the capacities of both plant and animal souls, as well as our own particular rational soul. Here I provide a simple diagram of the levels and capacities of the soul; however, the capacities are not in hierarchical order (Figure 20).

[33] Dunn, *Paul*, 78.

[34] There may still be scholarly disagreements here, and the literature is difficult to follow. See, e.g., Anthony Kenny, *Aquinas on Mind* (London and New York: Routledge, 1993).

Rational: will
active intellect
passive intellect

Animal: emotions
four internal senses:
sensus communis (common or unifying sense)
vis memorativa (sense memory)
vis aestimativa (instinctive judgment, estimative power)
phantasia (imagination)
five external senses:
sight, hearing, taste, touch, smell
appetites
locomotion

Vegetable: reproduction
growth
nutrition

Figure 20 Thomas Aquinas's list of the capacities of humans' tripartite soul

3.1.1 The vegetative soul and biology's verdict

The vegetative or plant soul, in the first instance, is simply the life principle. Our word "animate" comes from the Latin for soul, *anima*. However, the question of what makes something alive is now handled by biology. Biologists ask what the minimum requirements are for life. The basics are self-maintenance, growth, and reproduction. Thus, a sphere of proteins and other large molecules is living if, first, it has a membrane separating it from its environment; second, the membrane is permeable enough to allow for intake of nutrients; third, it has the ability to repair itself if damaged; and fourth, the ability to reproduce, even if only by splitting into two spheres, each of which grows large enough to split again.[35] The three functions

[35] The quest to discover how RNA-based reproduction could have first begun has been going on for decades. Nicholas Wade, "A Leading Mystery of Life's Origins is Seemingly Solved," *New York Times*, 14 May 2009: A15.

Thomas attributed to the vegetative soul were growth, nutrition, and reproduction. The one feature he failed to note was self-repair.

The physicalist thesis is that as we go up the hierarchy of increasingly complex organisms, all other capacities once attributed to the soul will turn out to be products of complex organization, rather than properties of a non-material entity.

3.1.2 The sensitive soul and subservient neural processes

The faculties Thomas attributed to the animal or sensitive soul were locomotion, appetite, sensation, and emotion. Let us consider some of these. Thomas's distinction between plant and animal on the basis of the ability to move from one place to another is still accepted.[36] The earliest and simplest form of locomotion is found in single-celled organisms equipped with flagella, whip-like structures on the surface of the cells that rotate and drive the cell forward. In humans and other higher animals, locomotion is controlled by a strip of cortex across the top of the brain, appropriately labeled the motor cortex.

Thomas distinguished two sorts of appetite. The sort we share with animals is that which is directed toward sensible objects such as food or mates. Sense-appetite includes a pleasure-seeking drive that inclines animals (and humans) to pursue what pleases their senses and avoid what hurts them, as well as for 11 kinds of emotion, such as love, hate, aversion, fear, despair, and anger.

In higher animals, appetite is mediated by the pleasure centers of the brain. Neuroscience shows these to be dependent on a balance of neurotransmitters—the chemicals that facilitate transmission of impulses from one nerve to another. Sexual desire is highly dependent on hormones. The hormone oxytocin is secreted by the posterior pituitary gland in mammals during sexual intercourse and breast feeding. It has been found to be a significant factor in pair bonding in animals and facilitates mother–infant bonding in humans.

A great deal of research has been done on the role of the brain and extended nervous system in sense perception. For example, visual perception in higher animals has developed from single, light-sensitive cells in primitive organisms. In humans and higher animals, signals are transmitted from light-sensitive cells in the retina via the optic nerve, through a series of processors, to the visual cortex, where additional circuits process it further, ultimately creating visual perceptions.

[36] Although Thomas's distinctions among plant, animal, and human capacities is still accepted, there are a few creatures that do not fit easily into one or another category. A lecturer once described a marine organism on the borderline between plant and animal. It spends most of its time attached to rocks, but in one phase of its life it develops a very simple brain, detaches from the rock, and moves to an area with more nutrients. Then it reattaches and consumes its own brain. The lecturer likened it to a professor who had gotten tenure.

The striking difference between lower and higher animals is that while the lower ones can respond to "visual" stimuli in their environments, such as light, they do so without knowing what they are doing—they lack consciousness. There is a phenomenon called blind-sight that helps clarify the difference between conscious and non-conscious perception. Certain victims of damage to the visual cortex are either completely blind or have blind spots in their visual fields. Nonetheless, they are receiving information about their environments. If asked to say where an object is they reply that they do not know; but if told to reach for it they do much better than would be expected by chance. So the value of consciousness is that we not only know things about our environment, but we also know *that* we know.

How consciousness arises from brain function is, as neuroscientists say, the hard problem. Since experts still do not agree on a theory of how this happens, I shall feel free to omit discussion of the current theories. However, it is interesting to note that scientists do not even agree on how far down the phylogenetic tree (of life) consciousness extends. Descartes is infamous for claiming that no subhuman animals had any sort of consciousness, but I do not think that anyone who has had a pet could agree. I have usually assumed that earthworms cannot feel pain, since they have no brain, but there seems to be no agreement as to whether the fish they are used to catch feel pain in their mouths.

Scientists would add at least one other sense: proprioception. This is the ability to detect one's orientation in space. Some religious experiences are thought to result from damping out the brain regions responsible for it.

In addition to the five "exterior" senses, Thomas postulated four "interior senses." These are particularly interesting in that they show Thomas's skill as a cognitive scientist, and also link up with quite detailed work in neuroscience. These are attributes shared with higher animals. Here is Timothy McDermott's (1926–2014) contemporary translation of Thomas's account:

> Higher animals must be aware of something not only when it is present to their senses but also in its absence, so that they can be prompted to seek it. So they not only need to receive, but also to retain, impressions of sense objects presently affecting them.

This ability to retain sense impressions in the absence of the stimulus is the interior sense called *phantasia* in Latin, and often translated as "imagination."

> In addition, animals need to be attracted and repelled not only by what pleases or displeases their senses but by what is useful or harmful in other ways: the straws birds collect must look good for nest-building. So animals must be able to perceive a significance in things that is not merely an externally perceptible quality. In addition to their particular senses . . . for receiving sense

> impressions and their imagination for storing them, animals must therefore have an instinctive judgment [the *vis aestimativa*; also translated as "estimative power"] . . . and a memory [*vis memorativa*, or "sense memory"] for storing those . . . Particular senses discern the particular sense-stimuli proper to them, but to distinguish white from sweet we need some common root sensitivity in which all sense-perceptions meet [the *sensus communis*—the "common" or "unifying sense"] . . .[37]

In this last sentence, Thomas is raising the issue that neuroscientists call the binding problem: how is it that we can put together the sound of a bark, the sight of brown hair, the sensation of soft fur, and the smell to recognize that we have before us a dog? This is one of the hardest problems in neuroscience next to consciousness itself.

An important question in neuroscience has been how the brain comes to recognize patterns. It is now believed that recognition tasks ordinarily depend on activation of large nets of neurons rather than on the firing of individual neurons. A "cell-assembly" is described as follows: "Any frequently repeated, particular stimulation will lead to the slow development of a 'cell-assembly,' a diffuse structure comprising cells . . . capable of acting briefly as a closed system."[38] This issue is clearly relevant to understanding Thomas's *phantasia* in that it is the reactivation of such an assembly that accounts for memory of the original set of stimuli.

Thomas's *vis aestimativa* is a particularly interesting faculty from the point of view of neuroscience. Joseph LeDoux (b.1949) is well known for investigating emotion. What he writes about "emotional appraisal" is relevant to distinguishing this estimative power from the *sensus communis*:

> When a certain region of the brain is damaged [namely, the temporal lobe], animals or humans lose the capacity to appraise the emotional significance of certain stimuli [but] without any loss in the capacity to perceive the stimuli as objects. The perceptual representation of an object and the evaluation of the significance of an object are separately processed in the brain. [In fact] the emotional meaning of a stimulus can begin to be appraised before the perceptual systems have fully processed the stimulus. It is, indeed, possible for your brain to know that something is good or bad before it knows exactly what it is.[39]

[37] St Thomas Aquinas, *Summa Theologiae: A Concise Translation*, ed. Timothy McDermott (Westminster, MD: Christian Classics, 1989), 121. Note that the interior senses must be involved in emotion, since judging and remembering what causes pleasure and pain are essential for the emotions involved in attraction and avoidance.

[38] Alwyn Scott, *The Controversial New Science of Consciousness* (New York: Springer-Verlag, 1995), 81.

[39] Joseph LeDoux, *The Emotional Brain: The Mysterious Underpinnings of Emotional Life* (New York: Simon & Schuster, 1996), 69.

So, using Thomas's terms, the *vis aestimativa* is a separate faculty from the *sensus communis*, and science shows that it works faster.

Thomas emphasized that the *vis aestimativa* is also capable of recognizing intentions. Neuroscientist Leslie Brothers (b.1953) describes the neural basis for such recognition in humans and animals. Social animals come equipped with neural systems that predispose them to pick out faces. The amygdala is necessary for interpreting facial expressions, direction of gaze, and tone of voice. Neurons in the same region respond to the sight of hands, and to leg motions typical of walking. Thus, there are particular neurons whose function is response to visual stimuli indicating intentions of other agents.[40]

LeDoux's research is also relevant to Thomas's *vis memorativa*, the ability to remember the emotional significance of a stimulus. A patient of Swiss physician Édouard Claparède had lost her abilities to create new memories as a result of brain damage. Each time Claparède (1873–1940) walked into the room, he had to reintroduce himself to her. He regularly held out his hand to greet her, but one day, when their hands met, she quickly pulled hers back; Claparède had concealed a tack in his palm and pricked her with it. The next time he returned she still had no recognition of him, but she refused to shake his hand, without being able to tell him why. He had come to signify danger; he had become a stimulus with a specific emotional meaning. She learned that Claparède's hand could cause her harm, and her brain used this stored information to prevent the unpleasantness from occurring again.[41] By investigating fear conditioning in rats, LeDoux has confirmed the crucial role of the amygdala, a distinctive cluster of neurons found in the anterior temporal lobe of each hemisphere, in developing this sort of memory.

3.1.3 The rational soul, still mysterious

For Thomas, the rational soul is what makes us distinctively human. He attributed to it two sorts of intellect, passive and active, and will. Passive intellect is a kind of memory—a memory of facts and ideas. It furnishes the mind with things about which to think. Neuroscientists now distinguish something like a dozen different memory systems. The two sorts of memory that Thomas distinguished are both classified as types of declarative memory and involve the medial temporal lobe of the brain. The *formation* of long-term memory requires the functioning of the hippocampus.

The active intellect is a power humans have, but not animals, of acquiring abstract information from sense experience and forming judgments. Not much is known about actual thought processes, because no brain is "wired" the same as

[40] Leslie Brothers, *Friday's Footprint: How Society Shapes the Human Mind* (New York: Oxford University Press, 1997), ch. 3.

[41] LeDoux, *Emotional Brain*, 180–1.

another. However, Anthony Kenny (b.1931) says that the intellect "is the capacity for acquiring linguistic and symbolic abilities."[42] While very little is understood about the complications of thinking, a great deal is known about the neural bases of language. Peter Hagoort (b.1954) describes the systems that are needed simply to recall and speak one word:

> All core steps in the speaking process are subserved by areas in the left hemisphere, which is the language dominant hemisphere in the large majority of people. Selecting the appropriate concept for speaking . . . seems to involve the left medial temporal gyrus. From there the activation spreads to Wernicke's area, which is pivotal in retrieving the phonological code of a word stored in memory. Wernicke's area plays a crucial role in the whole network of language processing by linking the lexical aspects of a word form to the widely distributed associations that define its meaning. This role is played by Wernicke's area in both language production and language comprehension. The lexical word form information is relayed to Broca's area in the left frontal cortex and/or the middle part of the superior temporal lobe in the left hemisphere. These areas play a role in the conversion of the phonological codes in memory into phonological words from which the abstract articulatory program is derived. In the final phase of preparing for articulation and execution of articulation sensorimotor areas become activated, with the possible additional contribution of the supplementary motor area and the cerebellum.[43]

So we can see the beginning of an understanding of the very complex brain processes that enable us to engage in language-based reasoning.

The third of Thomas's rational faculties was the will. This is the capacity to be attracted to goods of a non-sensory sort. As Kenny says, the will is "a power to have wants that only the intellect can frame . . . We can say roughly that the human will is the power to have those wants which only a language-user can have."[44] Along with intellect, this is the seat of moral capacities. Furthermore, since God is the ultimate good, the will also accounts for the capacity to be attracted to God.

Neuroscience now contributes to our understanding of both morality and religious experience. Antonio Damasio (b.1944) has studied the neural processes that

[42] Kenny, *Aquinas on Mind*, 18.

[43] Peter Hagoort, "The Uniquely Human Capacity for Language Communication: From *POPE* to [po:p] in Half a Second," in Robert J. Russell et al., eds., *Neuroscience and the Person: Scientific Perspectives on Divine Action* (Vatican City State and Berkeley, CA: Vatican Observatory and Center for Theology and the Natural Sciences, 1999), 45–56 (53, 55). His choice of the word "pope" was in honor of John Paul II, who had asked the Vatican Observatory to sponsor the series of conferences for which this paper was prepared.

[44] Kenny, *Aquinas on Mind*, 59.

go into practical reasoning, that is, the ability to make both moral and prudential judgments. In his book, *Descartes' Error*, he reports on cases in which brain damage has left cognitive functions, such as attention, perception, memory, reasoning, and language intact. Yet the subjects concerned suffered dramatic character changes afterwards. Hanna Damasio (b.1942) was able to determine from the damage to one such person's skull and from brain scans of others exactly which parts of the brain were destroyed—selected areas of the prefrontal cortices. The Damasios conclude that this area is "concerned specifically with unique human properties, among them the ability to anticipate the future and plan accordingly within a complex social environment; the sense of responsibility toward the self and others; and the ability to orchestrate one's survival deliberately, at the command of one's free will."[45] In short, what Thomas described as the "appetite for the good" appears to depend on localizable brain functions.

Brown and I published *Did My Neurons Make Me Do It?*,[46] in which we relied on the work of philosophical ethicist Alasdair MacIntyre, in his *Dependent Rational Animals*, for a *philosophical* definition of morally responsible action.[47] Our motto, summarizing MacIntyre's definition of the capacity for moral responsibility, is that it is *the ability to evaluate that which moves one to act in light of a concept of the good.* Note that his concern here is to ask the *philosophical* question of the essential requirements for attaining the capacity to act in a fully mature, rational, responsible, and moral manner. Here is how MacIntyre ties together the capacities that comprise moral reasoning:

> as a practical reasoner I have to be able to imagine different possible futures *for me*, to imagine myself moving forward from the starting point of the present in different directions. For different or alternative futures present me with different and alternative sets of goods to be achieved, with different possible modes of flourishing. And it is important that I should be able to envisage both nearer and more distant futures and to attach probabilities, even if only in a rough and ready way, to the future results of acting in one way rather than another. For this both knowledge and imagination are necessary.[48]

[45] Antonio Damasio, *Descartes' Error: Emotion, Reason, and the Human Brain* (New York: G. P. Putnam's Sons, 1994), 10.

[46] Nancey Murphy and Warren S. Brown, *Did My Neurons Make Me Do It? Philosophical and Neurobiological Perspectives on Moral Responsibility and Free Will* (Oxford: Oxford University Press, 2006).

[47] Alasdair MacIntyre, *Dependent Rational Animals: Why Human Beings Need the Virtues* (Chicago: Open Court, 1999).

[48] MacIntyre, *Dependent Rational Animals*, 74–5.

Brown and I determined from this overview what *cognitive capacities* are prerequisites for these abilities. We listed six general capacities:

1 A symbolic sense of self.
2 A sense of the narrative unity of life.
3 The ability to run behavioral scenarios and predict the outcome of various actions.
4 The ability to evaluate predicted outcomes in light of one's goals.
5 The ability to evaluate the goals themselves in light of abstract concepts of the good.
6 The ability to act in light of *1* through *5*.

I present only a sample of the neural prerequisites for some of these cognitive abilities. First, a symbolic sense (or concept) of self: This concept arises early in life, simply as the ability to distinguish one's body from the environment, but becomes more complex through maturation. LeDoux stresses the role of memory,[49] and it is also closely linked to a theory of mind; that is, the ability to attribute thoughts and feelings to other people. Shared neural circuits between actors and observers allow us to understand the actions and feelings of others as if they were our own, they prime us to act in similar ways and allow us to anticipate the same sorts of responses to our actions as we have seen in response to others' actions. We *share* their experiences rather than merely understanding them. The brain regions involved are the temporal lobe (more specifically, the middle temporal gyrus), the rostral inferior parietal lobule, and the ventral motor cortex.[50]

Another prerequisite for practical reasoning and moral responsibility is a narrative understanding of the history of one's life and recognition that the narrative will continue into the future, with alternative possibilities for the nature of that continuance. Understanding the narrative unity of our lives depends on the existence of a long-term memory for events from our pasts. This sort of memory is called episodic memory, which appears to develop early in childhood.

Symbolic language enhances mature humans' episodic memory, and also creates the capacity to anticipate stretches of time beyond the present and imaginatively to place ourselves in that future. It allows us to consider the distant consequences of our current actions. Thus, the third cognitive component needed for moral responsibility is the capacity to imagine different possible courses of

[49] Joseph LeDoux, *Synaptic Self: How our Brains Become Who We Are* (New York: Penguin, 2002), 9.
[50] Christian Keyser and Valeria Gassola, "Towards a Unifying Neural Theory of Social Cognition," in S. Anders et al., eds., *Understanding Emotions*, Progress in Brain Research vol. 156 (Amsterdam: Elsevier, 2006), 379–401, esp. 379.

action and predict their likely outcomes. Brown and I argue that mental capacities in general are best understood in terms of their role in guiding action. Thus, perceptions and beliefs are cognitive associative networks involving scenarios for potential actions.

Symbolic language is one of the most important sorts of external scaffolding for mental operations; that is, devices we can manipulate in order to increase our capacity to predict consequences of action. The limitless possibilities for combining linguistic elements in new ways enhances humans' abilities to construct an immense number of scenarios for possible future actions.

Sophisticated language allows not only for understanding a variety of abstract possibilities, but also for second-order evaluations of one's own evaluative processes. Brown and I call this the ability for self-transcendence. Humans, like all other organisms, are intrinsically active, and their actions are directed by biological drives and emotions, by immediate desires and longer-term goals, and by social expectations. The forms their action takes are shaped by learning, by imagination, by judgments about what will be effective. *Responsible* action is that which is consequent upon the ability to represent to oneself what one is doing and *why* one is doing it.

Recall that (following MacIntyre) I defined morally responsible action as action undertaken on the basis of evaluation of one's goals in light of *a concept of the good. Moral* responsibility, then, depends on the capacity to use moral *concepts* to describe (and *in so doing* to evaluate) one's own actions, character traits, dispositions, and so on. Consider the long history of culture required to possess abstract concepts such as these.

Finally, there is the question of what is required to take one's moral evaluations from the level of thought and put them into action. A long-standing interest in the philosophical literature is the problem of *akrasia*, that is, weakness of will. Why do we so often choose to do things that we know we will regret later?

Mark Hallett (b.1947) has developed a credible model to represent the normal relations among volition, action, and perception of agency, along with suggested neural correlates. Movement begins with motivation, and this leads to planning of a movement. He points out that it may be misleading to look for *an* initial event of willing since the brain is always working and producing actions; thus the relevant question is why the particular action that occurred was selected. When selected, the action can be executed. The perceptual component is alerted to upcoming movement from both planning and execution modules by feedforward signals.

Motivation is associated with limbic and prefrontal regions of the brain. Studies show that selection of the action to perform depends on different regions, depending on whether the choice is *which* action to choose (supplementary motor area, SMA) or *when* to move (dorsolateral prefrontal cortex, DLFPC). It is likely that movement is initiated in mesial motor areas and premotor cortex.

The movement command then goes to primary motor cortex. Corollary discharges appear to come from the SMA and dorsal premotor cortex to parietal areas, and these may be responsible for the sense of volition. Parietal and frontal areas maintain a relatively constant bidirectional communication. It is likely that this network of structures includes the insula. The sense of agency comes from the appropriate match of volition and movement feedback, likely also centered in the parietal area.[51]

This is but one account of brain regions involved in the initiation and completion of an action that one had judged to be moral. Section 3.2 will present a few instances of what is known about how brain functions interfere with proper human functioning, and this will add to an understanding of what needs to go right in order for us to act on our better intentions.

Readers may be thinking at this point that all of this discussion of the brain's relation to the capacities attributed to the soul has missed the most important issue: the soul is our capacity to interact with God. (Note that the issue of the soul's being the part of us that survives death has already been discussed under the heading of controversies regarding the intermediate state.)[52] A number of neuroscientists, however, have begun to study the role of the brain in religious experience. For example, Andrew Newberg (b.1966) has used brain scans to study Buddhist monks during meditation and Franciscan nuns during prayer. These scans show particular regions of the brain to be typically activated.

However, I omit detailed accounts of this research, first, because it takes account of too narrow a spectrum of religious experiences, namely, what spiritual directors call meditation or contemplation. Yet we could provide a long list of much more common activities and experiences that many would count as religious: petitionary prayer, communal worship, singing hymns, Scripture-reading, even pleasure felt while serving meals in a church's pantry. Second if one is a physicalist, then some region(s) or system(s) of the brain would need to be involved in any experience. And would brain involvement in, say, singing a hymn versus a folk song be different? Another field of study, the cognitive science of religion (CSR) is new. I shall report on it and offer a critique below (Chapter 10, sections 2.2 and 6).

[51] Mark Hallett, "Physiology of Volition," in Nancey Murphy, George F. R. Ellis, and Timothy O'Connor, eds., *Downward Causation and the Neurobiology of Free Will* (Berlin and Heidelberg: Springer-Verlag, 2009), ch. 7.

[52] There have been many claims that out-of-body experiences and near-death experiences provide empirical evidence for the survival of the soul into the next life. For the most inclusive and convincing argument against these arguments, I recommend Michael N. Marsh, *Out-of-Body and Near-Death Experiences: Brain-State Phenomena or Glimpses of Immortality?* (Oxford: Oxford University Press, 2010).

A final note: I have suggested in a number of my writings that adopting a physicalist rather than dualist anthropology could have important effects on church life. In some cases, it could counteract a tendency to focus on saving souls for the afterlife instead of a focus on discipleship in our own day. However, dualism has been combined with a strong emphasis on reforming the church in this world, as in the career of Teresa of Avila, who provided some of our most beautiful imagery of the soul, and at the same time reformed numerous Carmelite convents in Spain in the sixteenth century. For an excellent account of the consequences of physicalism for the church today, see Brown and Brad Strawn, *The Physical Nature of Christian Life*.[53]

3.2 Brain failure and human impairment

In this section, I shall present some neural causes of impairment, including criminality. These range from some of the most obvious to the more speculative.

We have already considered the work of the Damasios on the effects of damage to the orbital frontal area of the brain. Such cases involve personality or character change. Research suggests that damage to this same brain area also disrupts the ability to experience regret over past decisions and behavior.

A second case of gross brain damage that had clear effects on moral and criminal behavior is that of a 40-year-old man who inexplicably became a sexual impulsive with pedophilia. He had no prior history of sexual misconduct but began to frequent prostitutes and attempted to molest his stepdaughter. He was found guilty of child molestation and sentenced to prison. However, it was soon discovered that he had a large tumor pressing on his right orbital frontal cortex. When the tumor was removed, the patient's sexual impulsiveness diminished. But the impulsiveness later reappeared, and a brain scan revealed that the tumor had grown back. A second surgery again diminished the patient's sexual impulsiveness.[54]

The *New York Times* ran a report on neuroscientists who have found that food and recreational drugs have a common target in the "reward circuit of the brain," and that stress can cause biological changes, making humans and animals more susceptible to addiction. The author, Richard Friedman (b.1956), wrote that "this newspaper estimated that more than 59,000 Americans had died of drug overdoses in 2016."[55]

Stress and lower perceived social support correlate with fewer dopamine receptors, called D2s, in the brain's reward circuit. All rewards—sex, food, money,

[53] Warren S. Brown and Brad D. Strawn, *The Physical Nature of Christian Life: Neuroscience, Psychology, and the Church* (Cambridge: Cambridge University Press, 2012).

[54] Dean Mobbs et al., "Law, Responsibility, and the Brain," in Murphy et al., eds., *Neurobiology of Free Will*, ch. 15, p. 250.

[55] Richard A. Friedmann, "What Cookies and Meth Have in Common," *New York Times, Sunday Review*, 2 July 2017: 1, 4 (1).

drugs—cause release of dopamine, which conveys a sense of pleasure and helps to cement memory of the connection between the stimulus and the felt reward. The more D2s that people have, the higher their natural level of stimulation and pleasure. However, there is a vicious cycle: a low level of D2s is associated with lower activity in prefrontal cortex, which would impair judgment and the exercise of restraint. If this leads to use of drugs or overeating, these behaviors further lower the level of D2s.

Dean Mobbs (b.1975) and colleagues report that noninvasive structural brain imaging shows an 11 percent reduction in the amount of gray matter in the prefrontal cortex in patients with antisocial personality disorder. They report, further, that using brain imaging to look at function rather than mere structure reveals stronger relationships between brain and behavior. Using positron emission tomography, neuroscientists have found attenuated blood flow in the frontal lobes of violent individuals. In general, studies suggest that impulsive violent acts stem from diminished recruitment of the prefrontal cortex's *inhibition systems.*[56] Brown has suggested accordingly that studies of neuroscience and behavior should focus on what he calls "free won't" as much as on the traditional problem of free will.

Another area where damage may increase the propensity toward behaviors deemed to be criminal or antisocial is the amygdala. Imaging studies suggest a role of the amygdala in theory of mind, aggression, and the ability to recognize fear and sadness in faces. It has been theorized that perceived fear and sadness ordinarily inhibit aggressive behavior. The amygdala has been a major focus of attempts to understand the poor empathy and fear responses observed in psychopathic criminals. Mobbs hypothesizes that in psychopathic criminals the prefrontal–amygdala connections are disrupted, leading to deficits in contextual fear conditioning, regret, guilt, and affect regulation.

Mobbs also reports that Adrian Raine (b.1954) has used positron emission tomography to tease apart functional differences between premeditated psychopaths and impulsive affective murderers. Compared to controls, the impulsive murderers had reduced activation in the bilateral prefrontal cortex, while activity in the limbic structures was enhanced. Conversely, the predatory psychopaths had relatively normal prefrontal functioning, but increased right subcortical activity, which included the amygdala and hippocampus. These results suggest that predatory psychopaths are able to regulate their impulses, in contrast to impulsive murderers, who lack the prefrontal inhibitory machinery that stops them from committing violent transgressions.[57]

[56] Mobbs, "Law," 245.
[57] Mobbs, "Law," 247.

Just a few suggestive statistics: Studies show that 25 percent of defendants are medically and legally incompetent to stand trial. Antisocial disorder is ten times more common in prison populations than in the general population. Childhood trauma and mistreatment can actually stunt the development of the size of children's brains, and this in turn could be the cause of a large number of cases of antisocial personality disorder.

Mobbs concludes two things: First, these studies strongly suggest that some kinds of criminal behavior are associated with dysfunction of different regions of the brain. Second, though, given the role of environmental conditions, it would be inappropriate to expect full neurological localization of criminality.[58] I would add that finding the sorts of brain deficits described earlier does not provide much help in predicting or correcting antisocial behavior.

Again, I have only reported on a few cases of connections between neurobiological deficits or damage that seem to result in ineffective, or even evil behavior, but I hope to have provided enough examples to be convincing.

4 Philosophical issues

We saw (in section 2.2) that philosophical theories of human nature have had very significant influences on Christian anthropology. So I here return to the philosophical aspects of debates on human nature. Section 4.1 provides an overview of historical developments and, again, it is not intended to be a thorough history, but only to provide context for examining contemporary Christian physicalism. Much of this account will consist merely of reminders of relevant material presented previously.

Note that although I have continued to use the most common philosophical term for contemporary non-dualism, "nonreductive physicalism," there are other variants. I mentioned John Polkinghorne's "dual-aspect monism" and Kärkkäinen's proposal, with which I agree, that in Christian theology a more appropriate term would be "multi-aspect monism." Others are theologian Philip Clayton's "emergent monism" (not to be confused with William Hasker's (b.1935) "emergent dualism"); Christian philosophers Kevin Corcoran's (b.1964) and Lynne Rudder Baker's (1944–2017) "constitution view" according to which I am constituted by my physical body but am not identical with it; philosopher John Searle's (b.1932) "biological naturalism"; philosopher Donald Davidson's (1917–2003) "anomalous monism"; and there may be others. Note that "materialism" can be used as a term referring merely to human nature or to an entire worldview. Thus, "nonreductive materialism" is another possibility. "Monism" is another, and although I have used

[58] Mobbs, "Law," 252–3.

it to refer to biblical positions, it is ambiguous as to whether it is physicalist or Idealist monism (as H. Wheeler Robinson attributed to the OT).

In section 4.2, I focus on three significant philosophical problems that face the Christian physicalist. I do not pursue the opposite tack of exploring philosophical problems for dualists. These are well represented in modern philosophical literature. And, in fact, it might be said that the inability of some of the best minds, over a period of four centuries, to solve philosophical problems with dualism has been a major motivation for its rejection.

4.1 A sketch of modern and contemporary philosophy of mind

Since the mid twentieth century we have had a discipline called philosophy of mind; before that, pertinent issues regarding human nature were included in the philosophical discipline of epistemology.

We have already seen that atomism in early modern science forced a rethinking (for many) of the nature of the person. Descartes produced a radical substance dualism; the mind/soul is a substance whose essential nature is thinking, and thought comes in various modes: particular ideas, volitions, and passions. In his *Passions of the Soul* (1649), he grouped the passions with sensations, imagination, and perceptions that come from the external world. Now the mind is solely a function of consciousness, and many of its capacities in medieval thought were transferred to the body. And there are no longer animal or plant souls.

We have also seen that Hobbes took another obvious route: to count the whole of the human being as composed of particles. It would not be until later that theories such as his had important proponents and influences.

David Hume proposed that persons or selves *are* their minds, and that minds are merely bundles of perceptions. Minds are bound together only by the relations among the experiences, not by any existent entity that contains or unifies them.

Kant's sharp distinction between the phenomenal (that which was perceivable) and the noumenal (which included the self) provided a reconception of dualism that was as problematic as Descartes's when it came to explaining how the body or external world could causally affect the mind/self.

Even though Idealism could be called the most influential metaphysical stance of the nineteenth century, it is difficult to give a brief overview. T. L. S. Sprigge (1932–2007) writes: "Idealism is now usually understood in philosophy as the view that mind is the most basic reality and that the physical world exists only as an appearance to or expression of mind, or as somehow mental in its inner essence."[59] I shall make no attempt here to give an account of the various Idealists' theories of mind.

[59] T. L. S. Sprigge, "Idealism," in *Routledge*, 4:662–9 (662).

The Logical Positivist movement arose as a reaction against Idealism, especially that inspired by G. W. F. Hegel (1770–1831). It would be tempting to say simply that, if the goal is to eliminate the dualism of mind and matter, this was a return to materialism. However, due to the Cartesian notion that perception of the "external" world must be mediated by non-material ideas (or in positivist language, "sense data") in the mind, it did not (in most cases) represent as sharp a break with Idealism as was intended.

Neopositivist Ernest Nagel (1901–85) was a materialist, rejecting the existence of any non-material entities such as minds. However, we see here an early form of nonreductive materialism, in that he takes aesthetic qualities, ideals, suffering, and pleasure to be genuine aspects of experience, yet these are dependent on the temporal and spatial organization of bodies. It is here, I believe, that we begin to find true successors to Hobbes.

Behaviorism is a collection of twentieth-century positions, also aiming to escape from dualism, and perhaps best characterized as denying that mentalist sorts of words should be taken at face value as nouns with referents. Allow me to illustrate using an animal analogy. We would not ordinarily raise the question of whether a horse has a mind. We would certainly describe some horses as intelligent, but that does not imply that they have *something*, an intelligence. Gilbert Ryle (1900–76) and Ludwig Wittgenstein (1889–1951), I believe, have contributed the most in this regard.

In the 1960s and 1970s, the dualism–physicalism issue was expressed in terms of debates between mind–body dualists and mind–brain identity theorists. Arguments for and against dualism, for and against the identity thesis, appeared to be interminable. I believe the reason for this is that Anglo-American philosophical method was understood to be conceptual analysis. That is, in any statement (except tautologies) there were empirical elements (now ceded to science), but also conceptual elements. For example, the empirical question of how we acquire knowledge is empirical, but analysis of the *concept* knowledge was the philosophers' job. A correct analysis was thought to provide universal knowledge immune to scientific refutation. Thus, arguments for dualism were variations on arguments that the concepts *mind* or *mental*, and *brain* were not the same. This could be shown by giving examples of the nonsense that resulted from substituting the word "brain" in sentences where the word "mind" was at home. For example, if a person is seeing an orange-colored after-image it is "in the mind," but there is certainly nothing orange in the brain. So the mind cannot be identified with the brain.

J. J. C. Smart (1920–2012) considered eight arguments of this sort and provided rebuttals to each, thereby supporting the identity thesis. Regarding the example

above, he would not say that the after-image is a brain process, but that the experience of having it is a brain process, and the *experience* is not orange.[60]

However, W. V. O. Quine's demolition of the analytic–synthetic distinction made empirical knowledge once again relevant to philosophical arguments. In Jeffrey Stout's terms, conceptual *archaeology* needs to replace conceptual analysis. We need to know what a concept means in a particular era, and for a particular community of scholars (see Introduction and Chapter 2, section 6).

This view of philosophy entails that philosophical arguments based on current linguistic practices are suspect. Paul Feyerabend pointed out that common idioms are adapted to *beliefs*, not facts, and the acquisition of new knowledge (such as neuroscience) ought to be allowed to call such beliefs into question. Feyerabend says:

> It would seem to me that the task of philosophy, or of any enterprise interested in the advance rather than the embalming of knowledge, is to encourage the development of such new modes of approach, to participate in their improvement rather than to waste time in showing, what is obvious anyway, that they are different from the status quo.[61]

So the gradual displacement of conceptual analysis had already influenced a number of arguments for physicalism around 1970 on the basis of brain science. This by no means, however, indicates that there are no significant philosophical problems still to be solved for the physicalist.

4.2 Contemporary problems with Christian physicalism

Yet again, my coverage needs to be selective. What I shall focus on are philosophical problems that arise particularly for Christian physicalists. I shall ask, first, why physicalism should be thought to be true, or at least much more highly confirmed than dualism. Here I employ resources from philosophy of science. Second, there is the question of how to account for personal identity before and after the resurrection, even without a soul upon which to "hang one's identity." The third issue, which I am calling "the deal breaker," is that of reductionism. If there is no credible way to show it is *not* the case that higher human capacities are nothing but brain functions, then Christian faith is untenable.

There are other questions that could be addressed: What does it mean to be created in the image of God? For long periods in Christian history, this image was associated with the soul. Since Descartes's denying of souls to animals, the soul

[60] J. J. C. Smart, "Sensations and Brain Processes," *Philosophical Review* 68, no. 2 (April 1959): 141–56 (150–1).

[61] Paul K. Feyerabend, "Materialism and the Mind-Body Problem," in *Realism, Rationalism, and Scientific Method: Philosophical Papers, Volume 1* (Cambridge: Cambridge University Press, 1981), 161–75 (175).

was often counted as the main answer to what makes humans distinctive. If we have no souls, then in what does our distinctiveness lie? However, this chapter has already grown to an unwieldy length, and the three questions I do not cover are being addressed quite well by current theologians.

4.2.1 Epistemological evaluation

I have already discussed a version of Brown's criteria for judging the epistemological merits of physicalism: its consonance (or as Brown himself says, *resonance*) with Scripture, tradition, experience, science, and reason. I noted that this chapter is largely structured accordingly. I proposed that we might interpret "consonance with reason" here as absence of conflict with philosophy. So, in a sense, section 4.2 is my attempt (partially) to meet this criterion.[62]

I wrote above (Chapter 3, section 4.3) that, according to Imre Lakatos, the hard cores (central, unifying theories) of scientific research programs are often philosophical; for example, the atomism that served to structure and guide the development of early modern physics. In Lakatosian terms, it is certainly physicalism that serves as the hard core of the clusters of research programs that make up the cognitive neurosciences. (One exception here was the effort of Sir John Eccles (1903–97) to develop neuroscience on the basis of dualism, but nothing came of it.) So the positive heuristic (plan for continuing development of the program) is the ongoing attempt to explain additional instances of "mental" phenomena physicalistically.[63] Insofar as the physicalist program has been growing phenomenally over the past generation, and does not even have any competitors on the horizon, it is clearly the most highly confirmed, and thus the most rationally credible. I take this as the strongest possible scientific evidence for its truth.

However, my reason for reporting above on Lakatos's philosophy of science was not only to provide the backdrop to MacIntyre's understanding of tradition-constituted reasoning, but because I had found it valuable for understanding other systems of knowledge and therefore applicable to theological schools—theological research programs. So a project for the future might be to attempt to work out a "subprogram" in systematic theology centered on a physicalist anthropology.

4.2.2 Pre- and post-resurrection identity

Another problem for the physicalist is that of personal identity. In philosophical literature, the term "identity" is used in two ways. Numerical identity is distinguished from qualitative identity, and the former is at issue here: what are the

[62] Note that there are no sections corresponding to common human experience. This is because, just as language and concepts reflect the beliefs of the communities of which we are a part, so does our experience to some extent, and this is particularly true with something as subtle as our self-perceptions.

[63] I made this proposal in Brown et al., *Whatever Happened to the Soul?*, ch. 6.

criteria by which I am the same person now as I was 40 years ago, even though qualitatively I am quite different?

It is tempting to say simply that philosophers have not been able to solve the problem even for different stages in this life, so the inability of a Christian to account for pre- and post-resurrection identity cannot count as an obstacle to acceptance of a view that is so well supported otherwise. This is especially tempting given the fact that theological language presents more difficulties than does ordinary language, and our language especially breaks down when we come to matters of eschatology, such as the resurrection.

Nonetheless, I shall attempt to sketch a nonreductive physicalist account that is suitable for Christians. I shall conclude that it is not the body *qua* material object that constitutes our identities, but rather the higher capacities that it enables: consciousness and memory, moral character, interpersonal relations, and, especially, relationship with God.

David Wiggins (b.1933) has shown that to say "x is identical to y" requires the specification of a *covering concept*; one needs to be able to answer the question, "the same *what* as y?" Criteria of identity need to be tailored to fit the relevant covering concept. Consequently, it is necessary to ask what are the identity criteria for the covering concept *person* and to expect that these be different from identity criteria for a material object or even for a human body.[64]

There is a rich philosophical literature on personal identity. One long-standing argument is between those who stake personal identity on spatiotemporal continuity of the body and those who tie it to continuity of memories. It is a fact about us *in this life* that continuity of memory depends on temporal continuity of our brains and thus of the rest of our bodies. So these two criteria are essentially related. However, there is more to continuity of consciousness than a bundle of memories. This is the ability to recognize my conscious self over time, which is so unproblematic most of the time (e.g., when we wake up in the morning) that it may go unnoticed.

Recognition of oneself as oneself over time and after interruptions of conscious experience may have been presumed to be part of what philosophers have been referring to all along as the memory criterion, but I want to highlight it; I shall call this additional criterion the continuity-of-consciousness criterion.

I now want to argue that the combined memory–consciousness criterion is still too narrow, in that memory and continuity of consciousness together do not capture all of what we need in order to secure personal identity. Given the moral and social character of the kingdom of God, we need to add "same moral character" to our criterion. I propose that identity of persons depends as much on *character* identity as it does on memory/consciousness and bodily continuity.

[64] David Wiggins, *Identity and Spatio-Temporal Continuity* (Oxford: Clarendon Press, 1967), 1, 35–6, 50.

That is, a replica or transformed version of my body with all my memories intact would not be I unless she possessed my virtues (or vices), affections, and moral perceptions.

Another factor that needs to be taken into account in understanding personal identity is relationships. It is clear that a great deal of what makes us to be the persons that we are is our relations with other people, and I especially want to emphasize continuity of one's relations to others in the body of Christ and to Christ himself. Thus, I concur with those who emphasize that God's remembering, recognizing, and relating to me are essential to my post-resurrection identity.

What about bodily continuity? The nonreductive physicalist (or multi-aspect monist) assumption is that on this side of resurrection all of these other criteria are tied to the same body in the sense of same material object. But what if, given our faith in the resurrection, we revise our concept of personal identity accordingly, and define "my body" as that which provides the substrate for all of these personal characteristics? It is that which allows me to be recognized by others; that which bears my memories; and whose capacities, emotional reactions, and perceptions have been shaped by my moral actions and experience. It is these characteristics that make me who I am. So any body that manifests all of these is in fact Nancey Murphy.

This recognition allows us to avoid tortuous attempts, as in the early church, to reconcile resurrection with material continuity. Early theologians raised gruesome questions such as the problem of chain consumption: what if you die and a fish eats your body, then someone else eats the fish. Who gets the matter in the end? These attempts are based on failure to distinguish the covering concepts of *same person* and *same collection of particles.*

In some of my earlier writings, I have used the now popular concepts of *supervenience* and *realization.* "Supervenience" was first used as a technical term to describe the relation between moral properties and non-moral characteristics. The property of being a good person supervenes on properties such as being generous, truthful, etc. I argue (contrary to the most common usage) that the lower-level ("subvenient") properties *constitute* the higher-level properties, *only* given certain suitable conditions. For example, giving all one's money to the poor may *or may not* constitute goodness depending on whether one has a family to support. Supervenient properties are "multiply realizable." That is, there are a variety of different life patterns that constitute goodness.

So what if we say that this physical body *constitutes* me now,[65] but in the next life my personhood will be *realized* by a very different *kind* of body, yet one that still

[65] Recall that I mentioned the constitution theory of human nature as comparable to that of nonreductive physicalism (section 4 above).

embodies all of the characteristics I have mentioned above as necessary for personal identity? I am experimenting with language here, but I have already noted that the specification of identity criteria is difficult and controversial in all sorts of cases, and that eschatological language regularly falls short. Our language of bodies is all based on the physical features of this world, but we know that in the next world not just our bodies but all of creation will be as different as the resurrected Christ was from the bruised body laid in the tomb.

My earlier use of the concept of supervenience was in an attempt to evade reductionism with regard to humans, which is the problem to be addressed next.

4.2.3 Reductionism: The deal breaker

Throughout Christian history there have been both monist and dualist positions. For theological reasons, what we might call radical dualism such as Descartes's has been rejected in favor of "wholistic dualism": humans are essentially body and soul.

There is currently a monist position that Christians need to reject on theological grounds: reductive materialism. Otherwise, we lose all of the higher human capacities important to a theological vision of humankind. Recall that according to Dunn, Paul's anthropology emphasized our essential sociality, reflective thought, deep emotions and sustained motivations, and the ability to be touched by the Holy Spirit.

There are a number of reductive materialists in both philosophy and science. However, I report on just one, philosopher Daniel Dennett (b.1942) in his book, *Freedom Evolves*.[66] My account of Dennett's simulacra of human capacities does not line up in direct opposition to Paul's, but I believe it indicates well enough what the reductionist is willing to give up.

Many reviewers of Dennett's book accuse him of "bait and switch" arguments. Human sociality requires "intentionality," in the philosophical sense, including beliefs, desires, and intentions. Dennett claims we do *not* possess these characteristics. However, because attributing them to others allows us to predict behavior so well, this justifies us in taking "the intentional stance" toward others. Also, Dennett substitutes for altruism a concept of "benselfishness." Because of the social pay-off of being perceived as a good person, and the fact that the easiest way to maintain this perception is to actually be a good person, we behave so as to appear to be altruistic, but it is for selfish motives. This has been described as pseudo-altruism. Likewise, he has been accused of providing only for pseudo-responsibility and pseudo-freedom. His account of human thought is based on a view of language as nothing more than computer-like manipulation of strings of symbols. And Dennett, being a strident atheist, does not even think to provide an account of

[66] Daniel C. Dennett, *Freedom Evolves* (New York: Viking Press, 2003).

pseudo-spirituality. So the question for this subsection is how to account for the fact that we are not, as Dennett says, at the mercy of an organization of a trillion mindless robots.

My proposal is that the intellectual world is in the process of a rapid change in worldview or, in other words, a Kuhnian paradigm change across all of the sciences. This is the development of "complex (dynamical, self-organizing) systems theory." I shall begin with an account of why reductionism had such a hold on modern thought; however, my central purpose is an attempt to convey the sense in which complex systems theory contributes to a new postmodern worldview.

As already noted, in the mid seventeenth century, Hobbes developed a theory of human nature and society on analogy with atomism in physics. He was also the first to propose the concept of a hierarchy of sciences, reflecting a hierarchy of *complexity*. This is in sharp contrast to the ancient and medieval hierarchy of *beings*, and its significance here is that in the earlier hierarchy, all causation was from the top downward.

In the early days of modern physics, causal reductionism was inevitable. Following Epicurus, the essential assumption was that everything that happens is a consequence of the motions and combinations of atoms, whose behavior was deterministic. The atoms were not thought to be affected by the wholes they composed. Thus, the behavior of the wholes, from the levels of physics, to chemistry, to biology, psychology, and finally to social organizations, was ultimately the product of causation from the bottom up, and complex entities were *not* causes in their own right. So the defeat of causal reductionism in the human case needs to be the defeat of bottom-up causation; or as some would say, of part-to-whole determination.

The most significant criticisms of causal reductionism began with an emergentist movement in the 1920s. It was replaced by the reductionist ambitions of the positivists in philosophy of science. Exploration of a concept of *downward causation* began in the 1970s; and, currently, there are new and impressive attempts to define and defend both emergence and downward causation.[67]

Do these accounts of emergence and downward causation together solve the problems of the reduction of human capacities to biology? At first it may seem so, but then the question arises: How is it *not* the case that humans are, in a sense,

[67] In my judgment, the best account of downward causation is by Robert Van Gulick, "Reduction, Emergence, and Other Recent Options on the Mind/Body Problem: A Philosophic Overview," *Journal of Consciousness Studies* 8, nos. 9–10 (September/October, 2001): 1–34; and "Who's In Charge Here? and Who's Doing All the Work?" in John Heil and Alfred Mele, eds., *Mental Causation* (Oxford: Clarendon Press, 1995), 233–58. Both are condensed and reprinted in Nancey Murphy and William R. Stoeger, SJ, eds., *Evolution and Emergence: Systems, Organisms, Persons* (Oxford: Oxford University Press, 2007). On Emergence, see Terrence W. Deacon, "Three Levels of Emergent Phenomena," also in Murphy and Stoeger, *Evolution*, 88–110.

trapped by a combination of downward causation from our environments and the biological factors that do (still) contribute to our behavior? Where is there room for human agency? This is the point at which the resources of complex systems theory are required.

"Systems" thinking has been developing over the past half-century, although it has only recently begun to have a significant impact. Systems theory draws from a number of sources. There are roots in general systems theory, developed from the 1928 through the 1970s by thinkers such as Ludwig von Bertalanffy (1901–72). The idea was that the structure of complex entities, regardless of the academic field they fell into, could be modeled mathematically. Another early source was the study of cybernetics, the study of automated control systems, whether mechanical or biological. Current contributions come from information theory, nonlinear mathematics, the study of chaotic and self-organizing systems, and non-equilibrium thermodynamics.

Examples of the systems of interest range from autocatalytic chemical processes at the most basic, to weather patterns, insect colonies, social organizations, and, of course, human brains. Alwyn Scott (1931–2006), a specialist in nonlinear mathematics and neuroscience, describes nonlinear science as a meta-science, based on recognition of patterns in kinds of phenomena in diverse fields. This paradigm shift amounts to a new conception of the very nature of causality,[68] and he says that it in a way returns us to Aristotelian concepts of causation. Francis Heylighen (b.1960), professor at the Free University of Brussels, has made a bolder claim: systems theory provides the resources for an entirely new worldview, including ontology, epistemology, and ethics.[69] I attempt to set out here some of the essential concepts involved in this change.

Several authors call for a shift in "ontological emphases." Alicia Juarrero (b.1947) says that one has to give up the traditional Western philosophical bias in favor of *things*, with their intrinsic properties, for an appreciation of processes and relations.[70]

Systems have permeable boundaries, allowing for the transport of materials, energy, and information. The boundary is a matter of the tighter coupling of its components with one another relative to their coupling with entities outside of the system. The crucial components of complex systems are not things but *processes*. So, for example, from a systems perspective a mammal is composed of a circulatory system, a reproductive system, and so forth, *not* of carbon, hydrogen,

[68] Alwyn Scott, "The Development of Nonlinear Science," *Revista del Nuovo Cimento* 27, nos. 10–11 (2004): 1–115 (2).

[69] Francis Heylighen, unpublished paper written for a "Symposium on Research Across Boundaries," Luxembourg, June 16–19, 2010.

[70] Alicia Juarrero, *Dynamics in Action: Intentional Behavior as a Complex System* (Cambridge, MA: MIT Press, 1999), 124.

calcium. The organismic level of description is largely [illegible] level—that is, if the functional system works, it does not matter what its components are made of.

Systems are different from both mechanisms and aggregates in that the properties of the components themselves are dependent on their being parts of the system in question (e.g., a circulatory system can only *be* what it is within the context of an organism). Philosophers distinguish between internal and external relations. External relations do not affect the nature of the *relata*, but internal relations are partially constitutive of the characteristics of *relata*. An essential assumption of the predominant modern worldview was that the world is composed of *things* related to one another *externally*. Systems theory takes the relations among the constituent *processes* of a system to be *internal*.

Systems range from those exhibiting great stability to those that fluctuate wildly. This is because complex systems are nonlinear, that is, the current state affects the development of each future state. The difference in stability is due to the extent to which the system is sensitive to slight variations in initial conditions, and also to the extent to which there are feedback processes that either do or do not dampen out fluctuations. Systems at the extremes of this spectrum of stability are not of great interest to systems theoreticians. For example, a thermostatically controlled heating system is very stable but produces no novelty because it involves a negative feedback system that keeps the temperature within a set range. Imagine a "reverse" thermostat that provides positive feedback such that the hotter the building becomes, the more it increases the heating. This system is unpredictable, but not likely to last long.

Thus, the systems of interest are those in the middle of the spectrum. Chaotic systems are now familiar to many. They result from having a sensitivity to initial conditions that falls into a narrow range, resulting in their behavior falling into a predictable *range* of states.

More interesting are those at the edge of chaos. Here the system has the freedom to explore new possibilities and may "jump" to a new and higher form of organization. An understanding of how this can happen in terms of physics comes from the study of far-from-equilibrium thermodynamics.[71] Such systems are called complex *adaptive* systems. They are characterized by goal-directedness, at least insofar as they operate in order to maintain homeostasis. In the process of self-maintenance, they may create their own components. For example, in an autocatalytic reaction, molecule A catalyzes molecule B, which catalyzes more of A. The process will stabilize at some point unless additional materials are introduced into the system. In order for the system to maintain itself, the

[71] Ilya Prigogine and Isabelle Stengers, *The End of Certainty: Time, Chaos, and the New Laws of Nature* (New York: The Free Press, 1997).

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[56] Mobbs,
[57] Mobbs,

internal dynamics must determine which molecules are fit to be imported the system and survive.[72]

Complex adaptive systems theory has dramatic consequences for understa ing causation. While ordinary efficient causation is presupposed, systems th developed specifically because such causation is inadequate to describe com systems. This inadequacy is in part because complex systems operate on in mation as much as on energy and matter. More important is the fact that relations among the components of a system need to be thought of in term *constraints.* An efficient cause makes something happen. A constraint *reduce* number of things that can happen, due to the fact that the components are in nally related to one another such that a change in one automatically change other. Juarrero says that the concept of a constraint in science suggests "no external force that pushes, but a thing's connections to something else . . . as as to the setting in which the object is situated."[73] More generally, then, constr pertain to an object's connection with the environment or its embeddedne that environment. They are relational properties rather than primary qua in the object itself.

From information theory, Juarrero employs a distinction between context and context-sensitive constraints. For example, in successive throws of dice numbers that have come up previously do not constrain the probabilities the current throw; the constraints on the dice's behavior are context-free contrast, in a card game the constraints are context-sensitive: the chance say, drawing an ace at any point in the game are sensitive to history bec the rules of the game, the number of cards in the deck, and so forth create tions among the possible outcomes such that the probability of one occurr is related to all of the others. This account suggests that a better term in of "downward causation" is "whole–part constraint." The "higher-level" sys the whole, does not exert efficient, forceful causation on its components. Ra global features of the system are such that a change in one component cha the probabilities of the occurrence of other lower-level events.

Due to the role of probability in complex systems, it is necessary to do with the sharp distinction between determinism and indeterminism, either q tum indeterminacy or complete randomness. The appropriate middle ter "propensity," coined by Karl Popper to mean "an irregular or non-necessita causal disposition of an object or system to produce some result or effect."[74]

An understanding of the concept of a propensity has been aided by the st of nonlinear mathematics and especially chaotic systems. It begins with a vi

[72] Juarrero, *Dynamics*, 126.
[73] Juarrero, *Dynamics*, 132.
[74] See David Sapire, "Propensity," in *Camb. Dict.*, 657.

imaginary "state space" or "phase space," which is an imaginary "space" with me number of dimensions, in which a trajectory represents possible transitions m one state of the system to another. Chaotic systems theory introduced the cept of a "strange attractor" to describe the development of chaotic systems r time. This is a "shape" in phase space that depicts the boundaries within ich the system can be found during its evolution.

From the concept of a strange attractor, the idea of an "ontogenic landscape" been developed. This is a "topographical map" in which valleys represent areas phase space in which the system is likely to stay. Peaks represent states in which system will only be found as a result of a major perturbation, such as the injec- n of a great deal of energy. So the system has a propensity to remain within the leys. The topography represents a summation of the general effects of a vast mber of contextually constrained interactions among the system's component cesses.

Those who are not already familiar with material of this sort may be finding it ficult to follow. My mentor at Berkeley, Paul Feyerabend, once said that signif- nt conceptual changes begin with a group of people speaking nonsense to one ther, but it eventually makes sense, and a new paradigm or conceptual scheme reated. I myself believe that I am only now becoming fluent after a number of rs.

Now, as noted above, a number of us claim that systems thinking represents a nge in worldview. My late husband, James McClendon, and I developed cri- a to distinguish modern from postmodern theologies. Modern thought was strained by a foundationalist epistemology, a general theory of meaning as rence, and an individualist understanding of human nature.[75] We claimed that one who simply left all of the arguments relating to these issues behind had aped from the intellectual "space" of modernity. Epistemological holism left ndationalist arguments behind, the philosophical understanding of language erms of "meaning as use" left the modern dichotomy of referentialism versus ressivism behind.

While he correctly recognized the important role of individualism in modern- he claimed that arguments for and against it represented a "metaphysical" axis, I never knew why he called it that. It has only been in these past 20 years that me to see individualism as but one manifestation of the metaphysics of atom- and reductionism. And it is only now that I know the escape route from this dern set of arguments, namely complex systems theory.

Brown and I have argued that this set of new concepts, particularly those of nplexity theory, gives us the conceptual tools to explain how downward "causes"

[75] Nancey Murphy and James Wm. McClendon, Jr., "Distinguishing Modern and Postmodern Theologies," *Modern Theology* 5, no. 3 (April 1989): 191–214.

cause without violating the causal closure of the physical and without postulating causal overdetermination. Humans, who are complex, self-organizing, dynamical, adaptive systems, are partially decoupled from their biology, attend selectively to environmental constraints, and thus are able to become agents in their own right.[76]

However, higher animals possess these features as well. So a remaining question is what distinguishes adult humans' morally responsible actions from those of animals and even small children. See an overview of Brown's and my use of MacIntyre's account of moral agency in section 3.1.3 above. In short, morally responsible action depends on the ability to evaluate one's reasons for acting in light of a concept of the good.

So with this addition to the argument that organisms are (often) the causes of their own behavior—the argument I have made briefly in this subsection—I believe it is possible to make the claim to have eliminated many of the reductionist worries that seem to threaten our traditional theologically informed conception of ourselves.

5 Retrospect and prospect

This chapter has considered four sources that have contributed to Christian conceptions of the constitution and destiny of humans. Scripture is clearly, for many, the first source, but we have seen that later developments have often been read back into the texts. Thus, it is a fair question to ask whether current interests in monist accounts are having such an effect, or is it rather that, as with the cleaning of an old painting, the situation is more like the removal of varnish and smoke that long obscured it?

The theological tradition serves as another source of guidance, and here we find mixed messages. In the ancient period, contrasting philosophies contributed to contrasting theories of human nature. The Reformation provided powerful doctrinal pressures on Reformed and Catholic Christians to argue for substance dualism, while Anabaptists and Lutherans appear to have more options.

What we now call science has had a say in these matters. This is clearest in, first, the adoption of Aristotle's worldview, and then its replacement by atomism in modern science. Evolutionary biology, which is not covered here, drove Christians in two directions. One possibility was to say that if humans are so closely related to animals, and animals have no souls, maybe we do not have them either. The opposite direction has been the claim that it is all the more necessary to postulate the creation of a soul for each individual, in order to account for human distinctiveness among the animals. The question then arises, that given the long,

[76] Murphy and Brown, *Did My Neurons Make Me Do It?*.

continuous development of *Homo sapiens*, when should we postulate that God would have begun soul creation?

The science that most affects views of human nature at present is neuroscience. Its striking effectiveness in tracing processes and regions involved in all the capacities that Thomas attributed to the soul leads many to say that "it is the brain alone that does the work." But why should we believe that? Did not the speaker's brain simply cause him to make these noises?

It is essential, then, that one of the philosophical contributions is an argument (or explanation) to show that humans are not merely brain-driven organisms. I repeat here that it has taken a genuine worldview change, beyond the atomist–determinist–reductionist modern worldview; this change is rapidly appearing across all contemporary disciplines.

An explanation of post-resurrection identity that does not rely on a continually existing soul is another important issue. In section 4.2.1, I made the (currently) unusual move of relying on philosophy of science and neuroscience itself to address the question of physicalism, whose philosophical home now is the philosophy of mind.

The following (and last) chapter in a sense ties this text together. The question posed in Part I was whether there is an account of human reason sophisticated enough to evaluate the rationality of Christian thought. MacIntyre's account, I have argued, is the first so far. It involves showing that one's own tradition has had sufficient resources to solve its epistemological crises, and so Part II has examined some of the most significant crises presented by the modern worldview. MacIntyre also insists that self-justification does not end here. It requires opening one's enquiries to rival traditions and judging whether the most relevant rival(s) bring to it additional challenges that one's own tradition is unable to meet. Finally, can one show that one's own tradition has resources to offer the rival in order to solve the rival's own recognized crises, based on inconsistencies, features of the world it cannot explain, or simply its inability to progress further in its enquiries?

So Chapter 10 presents an account, as sympathetic an account as I am able, of the development and flourishing of a rival Naturalist Tradition, while noting the crises it has already faced or may recognize itself to face in the future.